

Contoso Infrastructure — M7 Motorway Extension Project Close-Out Report

Project Summary

Field	Details
Project Name	M7 Motorway Extension — Stage 3
Client	Transport for NSW
Contract Type	Design & Construct
Contract Value	\$385M (original), \$412M (final)
Duration	36 months (Jan 2023 – Dec 2025)
Location	Western Sydney, NSW
Scope	12km dual carriageway extension with 4 interchanges

Financial Performance

Budget vs Actual

Cost Category	Budget (\$M)	Actual (\$M)	Variance
Earthworks	48.2	52.8	+9.5%
Pavement & road base	35.6	34.1	-4.2%
Structures (bridges)	62.4	67.2	+7.7%
Drainage	18.3	17.9	-2.2%
Services relocation	22.1	28.6	+29.4%
Traffic management	14.8	16.2	+9.5%
Landscaping & finishing	8.2	7.8	-4.9%
Preliminaries & supervision	45.0	46.8	+4.0%
Subcontractors	95.4	102.6	+7.5%
Contingency (used)	35.0	38.0	—
TOTAL	385.0	412.0	+7.0%

Key Unit Rates Achieved

Item	Tendered Rate	Actual Rate	Variance
Bulk excavation (soil)	\$16.50/m ³	\$18.20/m ³	+10.3%
Bulk excavation (rock)	\$42.00/m ³	\$48.50/m ³	+15.5%
Concrete 40MPa (supply & pour)	\$295.00/m ³	\$310.00/m ³	+5.1%
Structural steel (erected)	\$5,800/t	\$6,100/t	+5.2%
Road base (150mm)	\$24.00/m ²	\$23.50/m ²	-2.1%
Asphalt wearing course	\$32.00/m ²	\$33.80/m ²	+5.6%

Earthworks Summary

Total earthworks volume: 1.8M m³

Type	Volume (m ³)	Rate Achieved	Notes
Cut to fill (on-site)	950,000	\$11.80/m ³	Better than estimate due to material quality
Rock excavation	280,000	\$48.50/m ³	40% more rock than geotechnical report predicted
Imported fill	420,000	\$30.50/m ³	Sourced from adjacent development (negotiated rate)
Unsuitable disposal	150,000	\$22.00/m ³	Contaminated material required off-site disposal

Key Issue: Geotechnical investigation underestimated rock volume by approximately 40%. This resulted in \$4.2M additional cost and 6-week program impact on the western cutting.

Lessons Learned

What Went Well

- 1. **Pavement works** — bulk purchasing agreement with quarry supplier saved 8% on aggregates
- 2. **Traffic management** — early engagement with TfNSW reduced stage changes by 15%
- 3. **Drainage** — precast elements reduced installation time by 25%

What Could Improve

- 1. **Geotechnical risk** — deeper investigation needed for rock profiles; consider additional boreholes on future motorway projects
- 2. **Services relocation** — unknown Telstra assets discovered 8 months into construction; budget additional 20% contingency for services on brownfield corridors
- 3. **Steel supply chain** — 12-week lead time blowout due to port congestion; secure early orders for structural steel on future projects

4. **Subcontractor management** — earthworks sub went into administration mid-project; maintain backup subcontractor agreements for critical trades

Recommendations for Future Projects

- Include minimum 15% rock contingency on any project with geotechnical uncertainty
- Mandate pre-construction services survey (DBYD + physical potholing) before earthworks commence
- Secure structural steel orders within 4 weeks of contract award
- Maintain panel of 3+ qualified earthworks subcontractors per region

This document is for workshop demonstration purposes only. All data is fictional.